

Python Programming Mini-Project Specification

Topic: Control Flow & Advanced Conditional Logic (If-Elif-Else)

Project Title:	Automated Tech Campus Multi-Tier Billing & Pass Engine
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1. Project Overview & Objectives

This project requires you to design and implement a comprehensive CLI-based system: **SmartCampus Automated Pass & Utility Tariff Calculator**. The system processes user demographics, access tiers, electricity usage, and peak hour surcharge variables to generate a consolidated itemized invoice.

Learning Objectives:

- Master multi-branch relational and logical decision pipelines (if / elif / else).
- Implement nested conditional statements to handle complex business rules.
- Validate input ranges and mitigate runtime exceptions using logical assertions.
- Apply range-based boundary checks and tiered pricing structures (slabs).
- Format formatted tabular output with proper currency and floating-point precision.

2. System Requirements & Problem Statement

The University Smart Campus Administration requires a unified python program to compute monthly access pass fees, parking permits, and hostel electricity surcharges for students and faculty etc. Your script must accept input data, execute validation checks, and apply rules dynamically based on explicit decision criteria.

Module A: User Classification & Base Pass Tariff

Base monthly access fee is determined by primary user classification and sub-category status:

User Category	Sub-Category / Condition	Base Fee (₹ / month)
Student	Undergraduate (UG)	₹ 500
Student	Postgraduate (PG)	₹ 350
Faculty / Staff	Resident Faculty	₹ 800
Faculty / Staff	Visiting / Guest Faculty	₹ 1,200

Module B: Surcharge & Discount Rules (Nested Logic)

Apply the following dynamic adjustments sequentially to the base pass fee:

Merit/Scholarship Discount: If user is a Student AND CGPA ≥ 8.5 , apply a 20% discount on the base fee. If CGPA is between 7.5 and 8.49, apply a 10% discount.

Seniority/Service Discount: If user is Faculty/Staff AND Years of Service > 10 , apply a 15% discount.

Vehicle Parking Permit (Optional Add-on):

- Two-Wheeler: Add ₹ 200/month.
- Four-Wheeler: Add ₹ 600/month. (If Four-Wheeler AND User is Student, add a peak-hour congestion surcharge of ₹ 150).
- None: Add ₹ 0.

Module C: Tiered Electricity Utility Billing (Slab Logic)

Calculate hostel/quarter electricity bill based on cumulative unit consumption using tiered slabs:

Consumption Slab (Units)	Rate per Unit (₹)	Fixed Service Charge
0 - 100 Units	₹ 3.00 / unit	₹ 50
101 - 300 Units	₹ 5.00 / unit (for units > 100)	₹ 100
301 - 500 Units	₹ 7.50 / unit (for units > 300)	₹ 150
Above 500 Units	₹ 10.00 / unit (for units > 500)	₹ 250

Module D: Input Validation & Exceptional Constraints

The system must enforce strict input integrity checks BEFORE processing calculations:

- CGPA must be floating-point value between 0.0 and 10.0.
- Electricity Units consumed cannot be negative.
- Years of Service must be non-negative integer.
- If invalid input is encountered, display an explicit error message (e.g., '[ERROR]: CGPA must be between 0.0 and 10.0') and terminate execution gracefully.

3. Expected Input & Output Format

Sample Interactive Execution Run (CLI Output Format)

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SMARTCAMPUS UTILITY & ACCESS PASS GENERATOR
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Select User Category (1: Student, 2: Faculty/Staff): 1
Enter Sub-Category (UG / PG): UG
Enter Student CGPA (0.0 - 10.0): 8.8
Select Parking Permit (0: None, 2: Two-Wheeler, 4: Four-Wheeler): 4
Enter Monthly Electricity Consumption (in kWh): 340

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                        CALCULATED INVOICE BREAKDOWN
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Base Access Pass Fee      : ₹ 500.00
Merit Discount (20%)     : - ₹ 100.00
Parking Fee (4-Wheeler)  : ₹ 600.00
Student Peak Surcharge   : ₹ 150.00
Net Pass & Parking Total  : ₹ 1,150.00

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Electricity Bill (340 kWh) : ₹ 1,900.00 (Slab calculated + Fixed Charge ₹150)

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TOTAL MONTHLY PAYABLE    : ₹ 3,050.00
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4. Submission Instructions & Rules

- File Naming Convention: Submission must be named as RollNo_Name_PythonProject.py (e.g., 22BCE1001_Rahul_PythonProject.py).
- Individual Work: This is an individual laboratory assessment. Code similarity above 15% will result in zero marks.
- Python Standards: Use Python 3.8+ features. Do NOT use external third-party utility libraries (only standard built-in functions allowed).
- Submission Deadline: Submit the .py file on the University LMS Portal before midnight on the specified due date.

--- End of Project Specification Sheet ---